

Mathematical Foundations for Computing

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Module 1

Module 1: Sets and Set Theory

▼

Module 2

Module 2: Number Systems and Bases

▲

Lesson 2.1 Introduction to Number Bases

- ✓

Introduction to number bases
Video • 13 min
- ✓

Introduction to number bases – conversion to decimal
Practice Assignment • Grade: 100%
- ✓

Conversion of integer numbers between different bases
Video • 7 min
- ✓

Conversion of non-integer numbers between bases
Video • 14 min
- ✓

Non-integer numbers conversion
Practice Assignment • Grade: 100%

Lesson 2.2 Operations with Binary Numbers

- ✓

Operations with binary numbers – addition
Video • 3 min
- ✓

Operations with binary numbers – subtraction
Video • 4 min
- ✓

Operations with binary numbers – multiplication
Video • 4 min
- ✓

Number representations
Video • 8 min
- ✓

Operations with binary numbers
Practice Assignment • Grade: 100%
- ✓

Why do we need to use these number conversions?
Dialogue • 15 min

Lesson 2.3 Summary and Assessments

- ✓

Topic 1 – looking back
Video • 2 min
- ✓

Check your understanding: End of module 2
Graded Assignment • Grade: 100%

Module 3

Module 3: Functions and Their Properties

▲

Lesson 3.1 Introduction to functions

- ✓

The definition of a function
Video • 11 min
- ✓

The definition of a function
Practice Assignment • Grade: 100%
- ✓

Plotting functions
Video • 9 min
- ✓

Plotting functions
Practice Assignment • Grade: 100%
- ✓

Injective and surjective functions
Video • 14 min
- ✓

Injective and surjective functions
Reading • 15 min
- ✓

Injective and surjective functions
Practice Assignment • Grade: 100%
- ✓

2D Graphing Calculator: Plotting some functions
Lab • 20 min

Lesson 3.2 More about Functions

- ✓

Function composition
Video • 4 min
- ✓

Function composition
Practice Assignment • Grade: 100%
- ✓

Bijjective functions
Video • 11 min
- ✓

Bijjective functions
Practice Assignment • Grade: 100%

Module 3 summary

0:13 / 3:10

2x





Studying functions provides students with skills and knowledge that are essential for understanding and applying mathematical concepts in various contexts. Functions are widely used in programming. Functions in programming are modular units of code designed to perform specific tasks. They encapsulate a set of instructions, allowing for code reuse and organisation. Here are some key learning concepts of this topic:

1. **Definition and notation:**
- Understand the definition of a function: A function f maps one element from a set A to a unique element in a set B , we write $f : A \rightarrow B$.

Understand the domain, the co-domain, and the range of a function.
2. **Properties of functions:**
- Understand and be able to show that a function is Injective(one-to-one), surjective(onto) or bijective.
3. **Inverse function:**
- Identify and find the inverse of a function, provided it exists, and understand the relationship between a function f and its inverse f^{-1} .
4. **Plotting a graph of a function:**
- You need to know how to plot a graph of a given function f and use this graph to show if the function f is injective, surjective.

If a function f is a bijection than you need to understand the relationship between the graph of f and that of its inverse, f^{-1} .The graphs of f and f^{-1} are symmetric with respect to the straight line $y = x$

5. **Function Composition:**

Understand and perform function composition, denoted as $(f \circ g) = f(g())$

6. **Specific Functions:**

Understand linear, quadratic, ceiling, floor, exponential and logarithmic functions.
- As you progress through your course, it is crucial to regularly assess your understanding and capabilities against the learning outcomes. This exercise is designed to help you reflect on your learning journey, identify areas where you may need to deepen your knowledge or skills, and develop a plan for improvement. Please follow the steps below:
1. **Review the learning outcomes.** For each of this topic’s learning outcomes, rate your understanding on a scale of 1-5 (1 = very low, 5 = very high). Provide a brief explanation for your rating.

2. **Identify areas for improvement.** Review your ratings and explanations:

a. Identify any areas where you rated your understanding below a 4.

b. For each area identified, describe specific aspects you find challenging or unclear.

3. **Create an action plan.** For each identified knowledge gap, outline specific actions you will take to improve your understanding. Consider the following strategies:

a. Reviewing the course materials and textbooks.

b. Seek out additional resources that could aid your understanding.

c. Practicing additional problems or exercises.

d. Seeking help from the instructor or peers.

e. Scheduling additional study sessions on the topic.

Implement your action plan and regularly review your progress. Adjust your strategies as needed based on your ongoing self-assessment and any new feedback you receive. Remember, learning is a continuous process, and it’s okay to revise your plan to better suit your needs.

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✓ Completed

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